

2.4.37

EE25BTECH11041 - Naman Kumar

Question:

Find a vector of magnitude 6, which is perpendicular to both the vectors $2\hat{i} - \hat{j} + 2\hat{k}$ and $4\hat{i} - \hat{j} + 3\hat{k}$

Solution:

Given Vectors

$$\mathbf{A} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}, \mathbf{B} = \begin{pmatrix} 4 \\ -1 \\ 3 \end{pmatrix} \quad (1)$$

Let required vector be,

$$\mathbf{C} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \quad (2)$$

Using Inner Product,

$$\mathbf{C}^T \cdot \mathbf{A} = 0 \text{ and } \mathbf{C}^T \cdot \mathbf{B} = 0 \quad (3)$$

$$\mathbf{C}^T \cdot \mathbf{A} = 2x - y + 2z = 0 \quad (4)$$

$$\mathbf{C}^T \cdot \mathbf{B} = 4x - y + 3z = 0 \quad (5)$$

Using Rank to Analyze the system

$$2x - y + 2z, 4x - y + 3z \quad (6)$$

$$\mathbf{A} = \begin{pmatrix} 2 & -1 & 2 \\ 4 & -1 & 3 \end{pmatrix} \quad (7)$$

Using Row Transformations to Get Row Reduced echelon Form

$$\mathbf{A} = \begin{pmatrix} 2 & -1 & 2 \\ 4 & -1 & 3 \end{pmatrix} \xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 2 & -1 & 2 \\ 0 & 1 & -1 \end{pmatrix} \quad (8)$$

$$\xrightarrow{R_1 \rightarrow \frac{1}{2}R_1} \begin{pmatrix} 1 & -\frac{1}{2} & 1 \\ 0 & 1 & -1 \end{pmatrix} \quad (9)$$

$$\xrightarrow{R_1 \rightarrow R_1 + \frac{1}{2}R_2} \begin{pmatrix} 1 & 0 & \frac{1}{2} \\ 0 & 1 & -1 \end{pmatrix} \quad (10)$$

$$\mathbf{A} = (\mathbf{IX}), \mathbf{I} \text{ is identity matrix} \quad (11)$$

And, $\mathbf{X}$ is

$$\begin{pmatrix} \frac{1}{2} \\ -1 \end{pmatrix} \quad (12)$$

Since rank of matrix is $2(\leq 3)$, there are infinite many solutions $R^3 \rightarrow R^2$

From the Row Reduced Echelon form(RREF),we can write the new system of equation:

$$x + \frac{1}{2}z = 0 \quad (13)$$

$$y - z = 0 \quad (14)$$

Therefore vector **C** using equations (13) and (14) is

$$\mathbf{C} = (x, -2x, -2x) = x(1, -2, -2) \quad (15)$$

Now getting vector with magnitude 6

$$|C| = 6 \quad (16)$$

$$|x(1, -2, -2)| = 6 \quad (17)$$

$$|x| \sqrt{(1)^2 + (-2)^2 + (-2)^2} = 6 \quad (18)$$

$$|x| \sqrt{1 + 4 + 4} = 6 \quad (19)$$

$$|x| \sqrt{9} = 6 \quad (20)$$

$$|x| = 2 \quad (21)$$

Therefore final vectors are

$$C_1 = \begin{pmatrix} -2 \\ 4 \\ 4 \end{pmatrix}, C_2 = \begin{pmatrix} 2 \\ -4 \\ -4 \end{pmatrix} \quad (22)$$

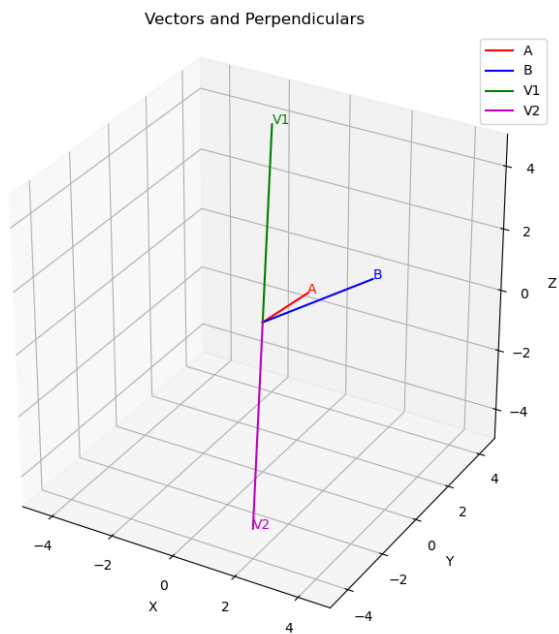


Fig. 1